

Polymorphs display different solid-phase physical properties

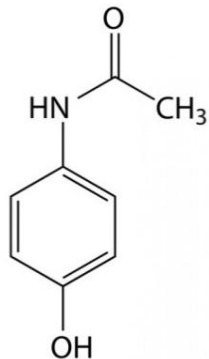
➤ polymorphism significantly affects properties of the solid form of a drug:

- solubility
- dissolution rate
- bioavailability/bioequivalence
- chemical stability ...

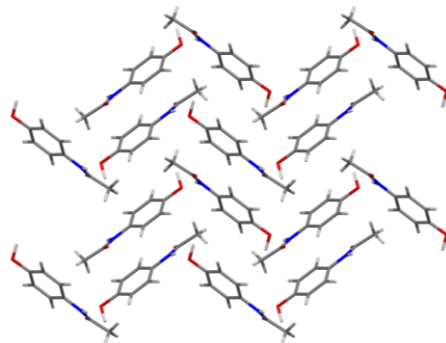
Polymorphism of Acetaminophen

Paracetamol

- an analgesic drug

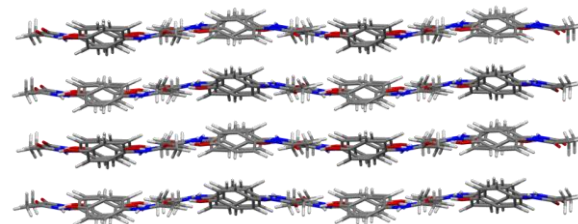


Form I (P2₁/a)



- thermodynamic polymorph
 - commercially sold
 - herringbone arrangement
 - tableting problems
 - no direct compression
 - ➔ binding agents
- costly in time and materials

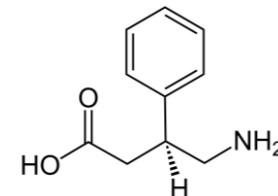
Form II (Pcab)



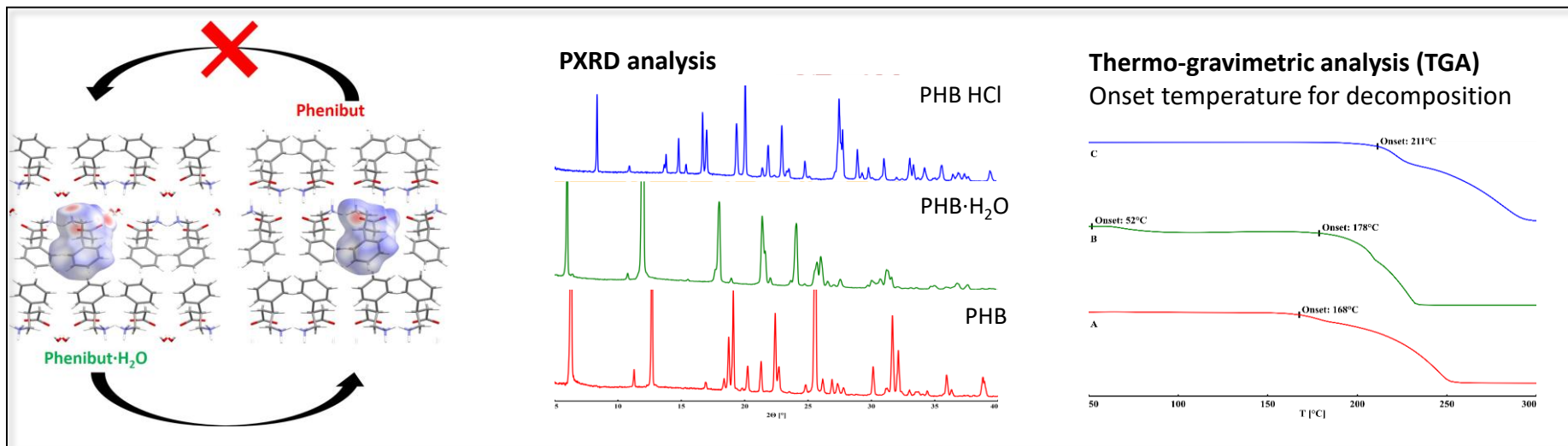
- kinetic, metastable polymorph
 - well-developed slip planes
 - plastic deformation upon compaction
 - direct tableting
 - slightly more soluble than form I
- processing advantages & commercial benefit

Form II is better! Why is it still not on the market?

Crystal packing & phase stability studies



- anxiolytic drug phenibut vs. monohydrate
- family of γ -amino butyric acid (GABA)

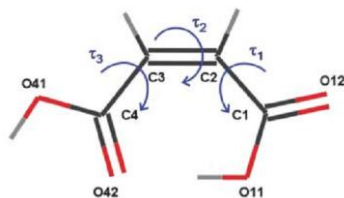


Structure-property relation

- calculations of intermolecular interactions energies
- conversion of PHB·H₂O to PHB → substantial gain in energetically favorable interactions for each phenibut molecule
- gain $\Delta E > 200$ kJ/mol

Latent polymorphism and disappearing polymorphs

→ crystal forms that fail to reappear after their initial isolation

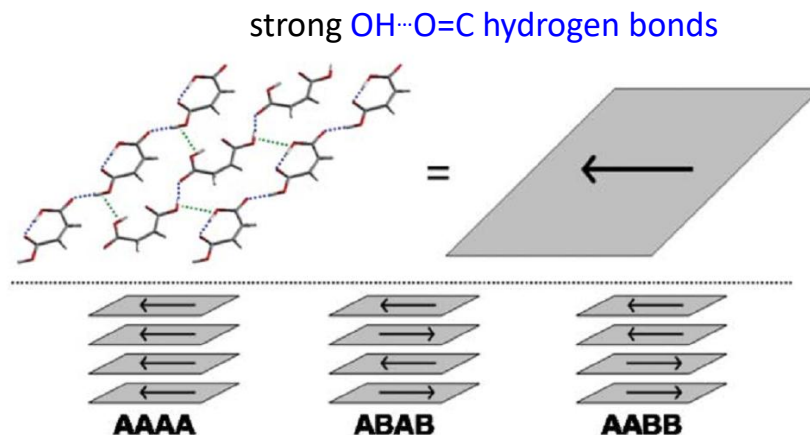


maleic acid

Bulk product of the chemical industry

- > 1000 tonnes per year
- salt-forming agent in the pharmaceutical industry

- crystals known and studied since 1881
- X-ray analysis in 1925 1939, 1952 and 1974
 - same crystal form $P2_1/c$
- 2006 a co-crystallisation attempt
 - **new polymorph** Pc
 - **not reproducible**



stacking of the **molecular sheets** in
observed and predicted crystal structures

- Computation of stable polymorphs – 16 lowest energy structures
- Crystal packing analysis

Disappearing polymorphs: Ranitidine·HCl block buster

- H2-antihistaminikum
- anti-ulcer drug
- prevent and relieve heartburn



1977: synthesis, 1st rational drug design

1978: “plate patent” of **form I** (valid till 1995)

- **Zantac**: introduced, licensed and marketed

Some time later...

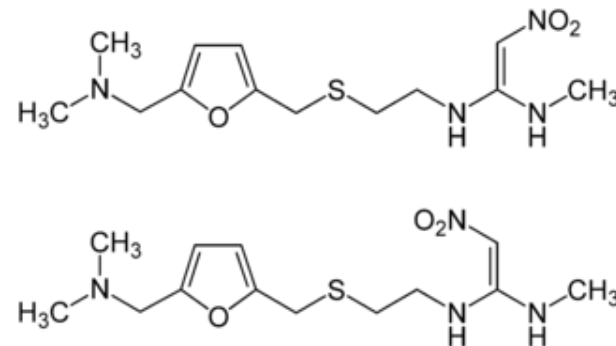
in the 13th production charge:

→ new „needle“ **form II**

Pure phase I is not producible at the Glaxo factory since then

1985: new “needle patent” of form II

- marketing



Ranitidine·HCl

a patent protection and extension block buster

Early 90-s: the sales reached 3.8 billion \$/year =
50 % of whole merit of Glaxo group



New player:

a generic pharma company **Novopharm**:

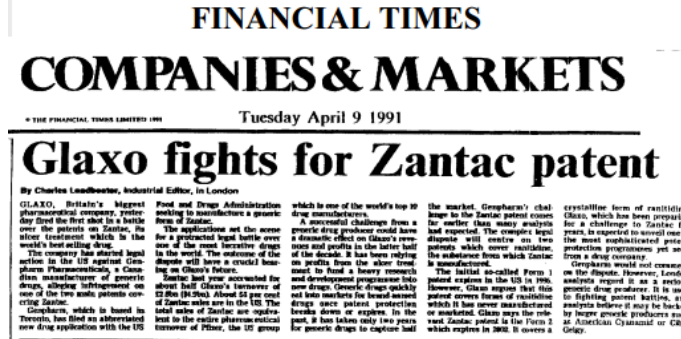
- could not reproduce the patented form I
- claim „needle patent“ to be invalid

➤ Novopharm could market form II

Glaxo sue Novopharm and defends:

- an **independent lab** can **reproduce** the protected procedure of **phase I** formation

Novopharm lost the first battle!



Ranitidine·HCl

a patent protection and extension block buster

The show goes on...

1994: **Novopharm** – new preparation method for the vanished **pure form I**

- wanted: **new patent and marketing**

Glaxo sue Novopharm again and claim:

- form I is not pure
- contains form II (protected by the „needle patent“)

Novopharm proves the purity of form I

- market a genericum Novo-Ranitidine

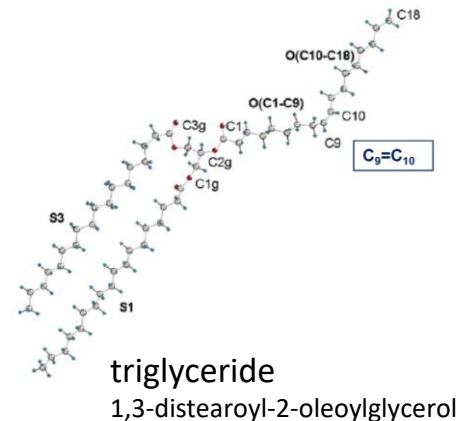
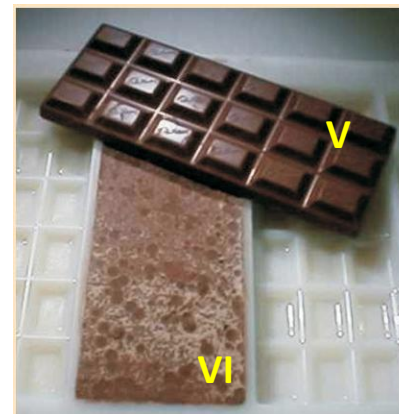


First major legal event!

Brought scientific field of organic polymorphism into
the glare of commercial and industrial activities

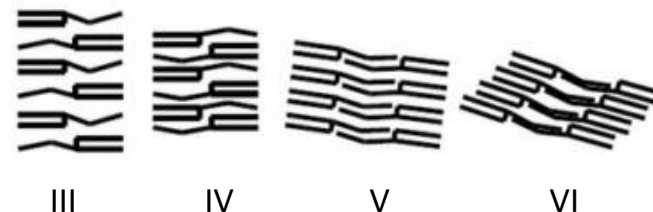
Six polymorphs:

- | | | |
|-----|--|--------------|
| I | rapid cooling of the melt | m.p. 17.3 °C |
| II | rapid cooling of the melt with 2 °C/min | m.p. 23.3 °C |
| III | crystallization at 5-10 °C | m.p. 25.5 °C |
| IV | crystallization at 16-21 °C | m.p. 27.3 °C |
| V | slow crystallization | m.p. 33.8 °C |
| | ⇒ desired edible form | |
| VI | from form V after several months at RT | m.p. 36.3 °C |
| | ⇒ thermodynamic stable form; grainy, dull surface, does not melt in the mouth | |



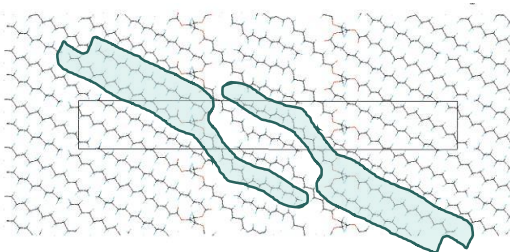
Crucial processing conditions:

- temperature control
- cooling/heating rate
- seeding



Production of form V

1. melting at 50 °C
2. cooling the melt to 22 °C with 1 °C/min
3. hold at 22 °C for several minutes
→ formation of a number of crystal nuclei of form II and III
4. warm up to 31 °C with 4 °C/min
→ form IV melts
5. cooling
→ conditions depend on properties of desired product

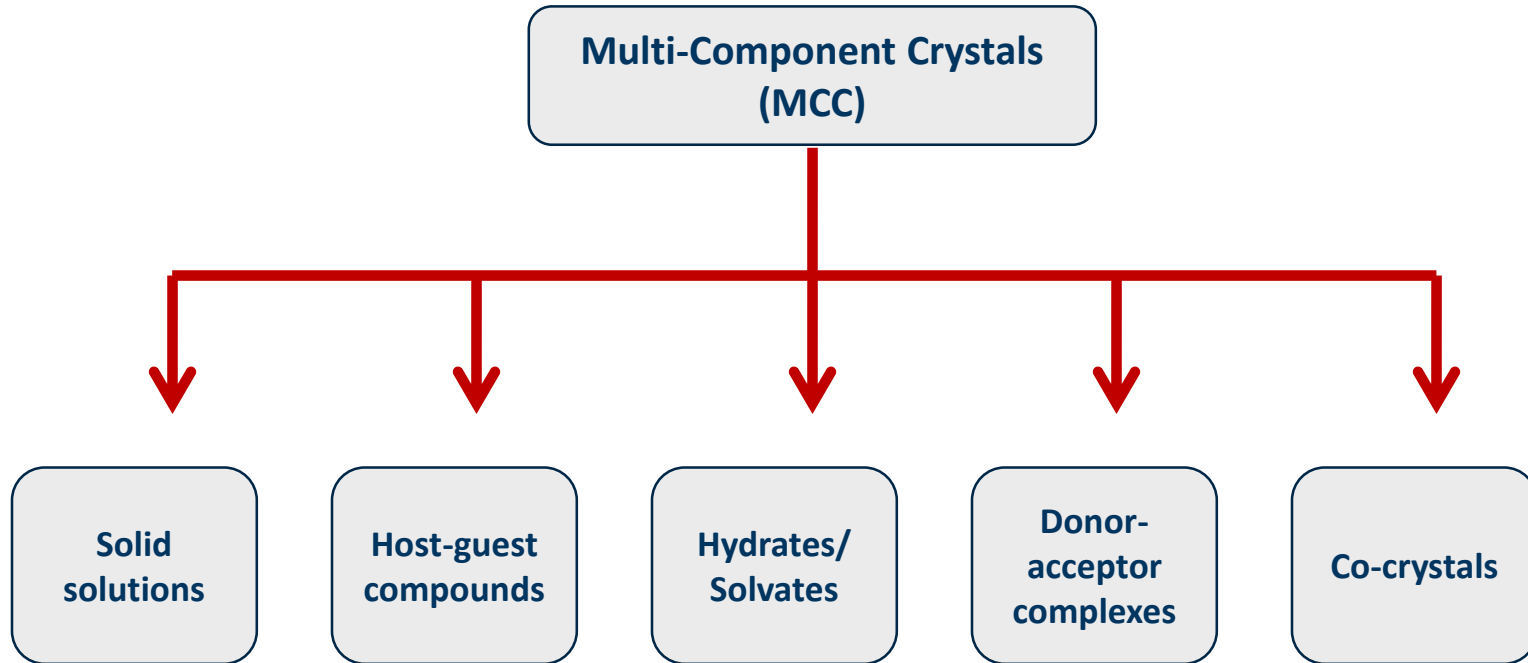


Search for better chocolate formulations
> **heat-resistant**: marketing potential

- 2016 Barry Callebaut
- new chocolate form – patent
- melting process starts 4 min
- m.p. 38 °C - **four degrees higher** than conventional chocolate
- production process and exact formulation are classified

Advantages:

- slower melting
- sticks less to the packaging,
- no need to be refrigerated
- easier to sell in warmer regions
- opens up opportunities for mail-order sales



Similarities:

equal laws of crystal packing
same fundamental level

- **maximize density and minimize free volume** - close packing
- **gain energy** - Minimize repulsive interactions & maximize attractive interactions
- **satisfy HB donors and acceptors and any other special kind of intermolecular interactions**
structure determining (the strongest HB donors always connect to the strongest HB acceptors)

Differences:

crystal growth > 1 chemical compound involved
consideration of chemico-physical properties
relative solubilities, vapor pressure, intrinsic chemical nature...

Multi-Component Crystals (A_mB_n)

why do some crystals contain more than one chemical component?

isolable crystals = free energy minima ΔG

In MCC: $\Delta G (A_mB_n) < \Delta G (A \text{ or } B)$

Reason for A_mB_n free energy minimisation:

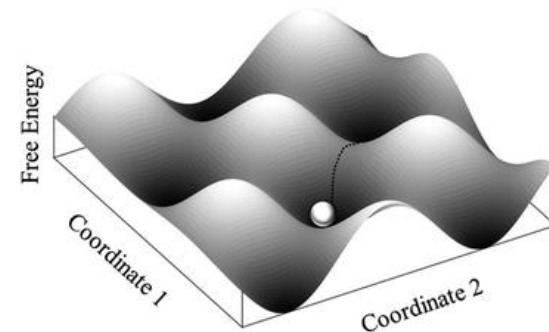
➤ **enthalpy**

- ⇒ intermolecular interactions $A \cdots B$ is more favourable than $A \cdots A$ or $B \cdots B$
- ⇒ definite composition & A:B stoichiometry (1:1, 1:2, 2:3)

➤ **entropy**

- ⇒ molecules of the same size, shape and chemical nature
- ⇒ variable stoichiometry A_mB_n - m and n from 0 to 1

Free energy surface of a landscape

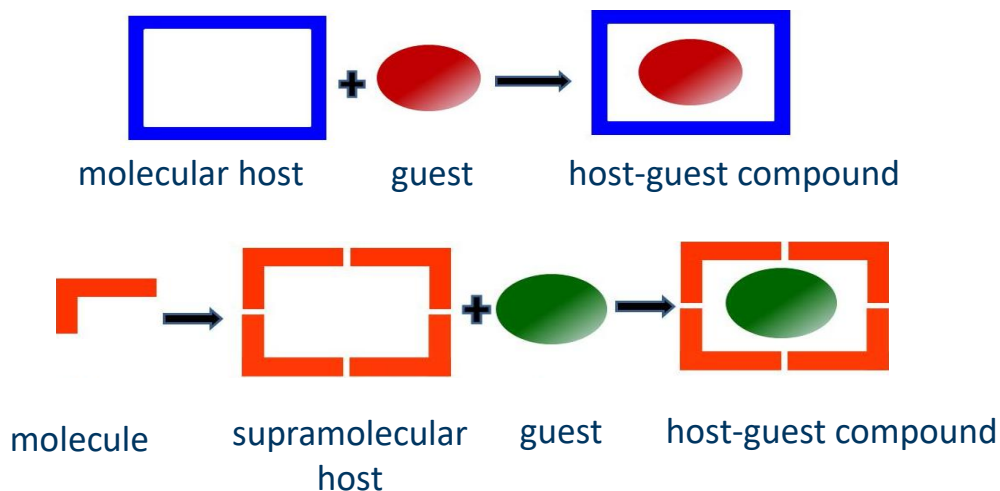


Molecular host: a single molecule/polymer which has a cavity or cleft

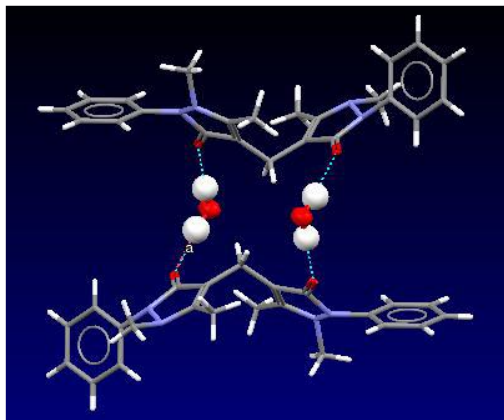
- host structure is inherently stable (similar to the behavior of zeolites) – (COFs/MOFs)

Supramolecular host: stabilized through non-covalent interactions

- host framework is stable only in the presence of guest $\Rightarrow$ Clathrates, HOFs



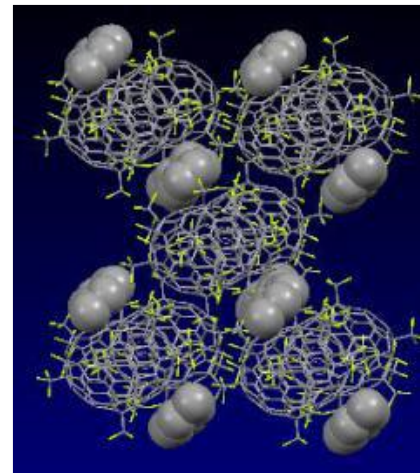
Polar solvents: stabilized in the crystal structure through directional interactions



4,4'-Methylene-diantipyrine monohydrate

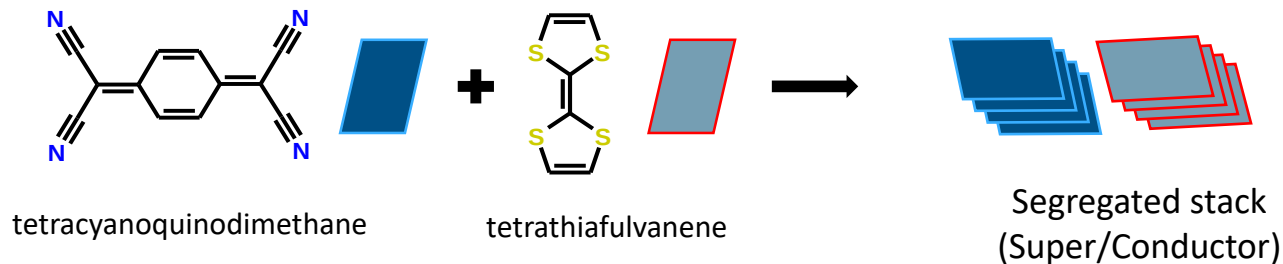
- stabilize the crystal packing
- competing reaction to formation of other MCCs

Non-polar solvents: fulfill a spacefilling role – fill voids and cavities in the crystal packing

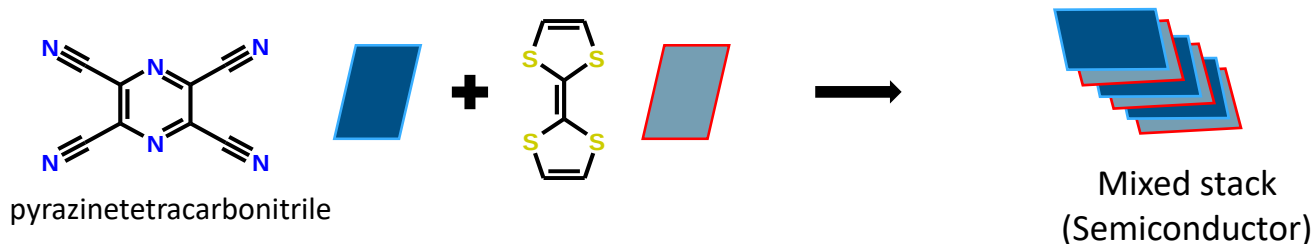


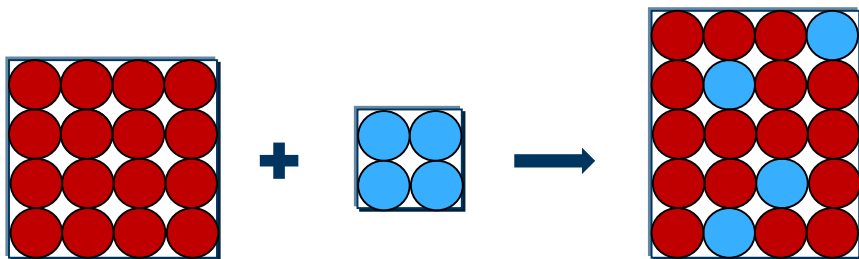
F-substituted fullerene & toluene

Molecular conductors: TCNQ-TTF – first organic metal



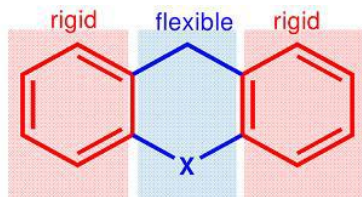
Molecular semiconductors/insulators: TCP-TTF





random mixture of
solute in „solvent“

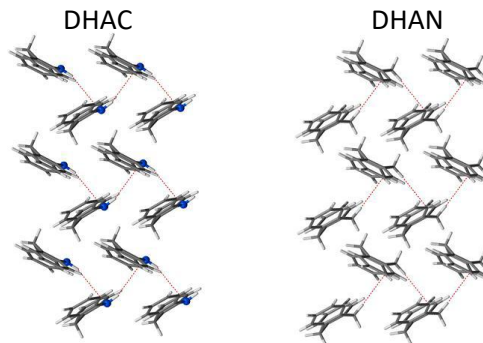
- atoms, ions or molecules of the same size and shape
- driving force - gain in entropy
- often inorganic chemistry (brass, bronze, bell material)
- less common among organic solids (size and shape mismatch)
- rarely > 2 components in solid solution



$X = CH_2, NH$

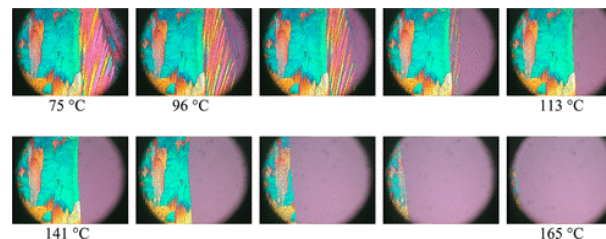
$X = CH_2$, dihydroanthracene (DHAN)

$X = NH$, dihydroacridine (DHAC)



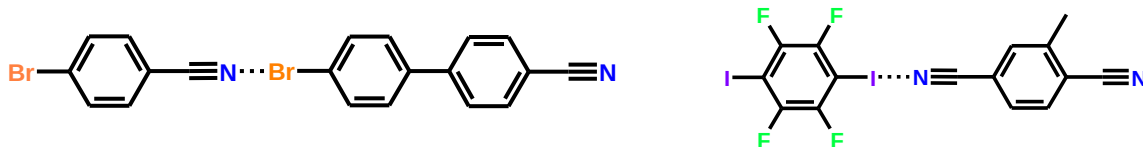
Hot stage microscopy

DHAC DHAN

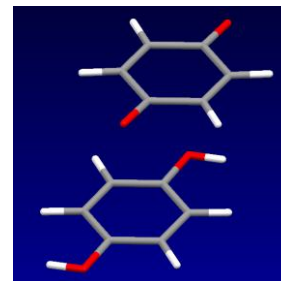
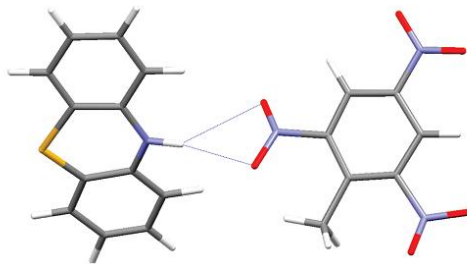
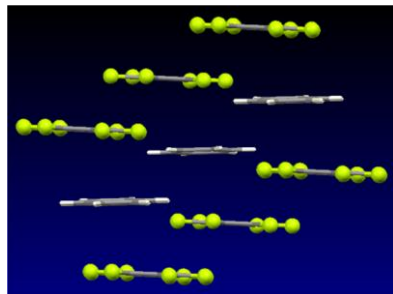
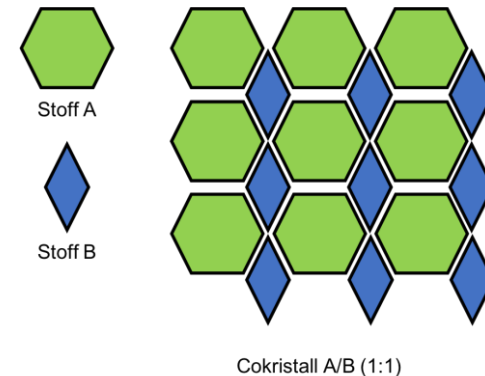


continuous melting process => formation of a SoSo within the contact transition region

Cocrystals / co-crystals



... are solids that are crystalline single phase materials composed of two or more different molecular and/or ionic compounds generally in a stoichiometric ratio.



Co-crystal:
multicomponent
ordered structure
defined position
defined number
new compound

Co-crystal Engineering with 2,4,6-Trinitrotoluene

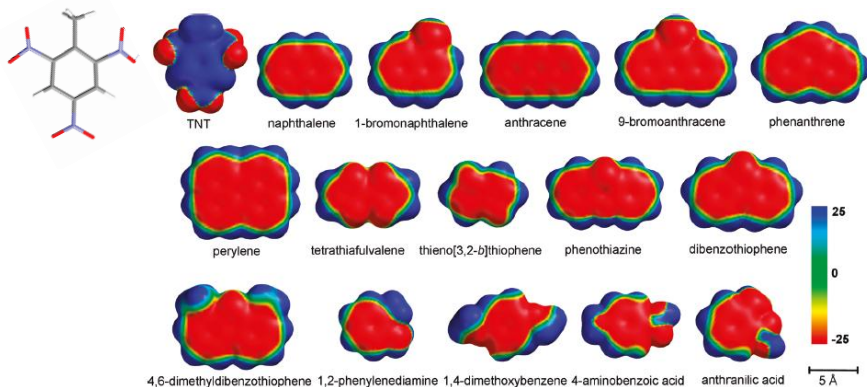


Figure 1. Electrostatic potential surfaces of TNT and the cocrystal formers calculated using the semiempirical method and the AM1 model. Red and blue surfaces represent electron rich and poor regions, respectively. Surfaces are normalized between -25 and 25 kJ/mol.

Adressed properties:

- melting point
 - packing coefficient
 - decomposition energies
- controlled explosion

Monocoordinated amine-nitro interactions
in TNT/1,2-phenylenediamine co-crystal
→ zig-zig chains generate sheets

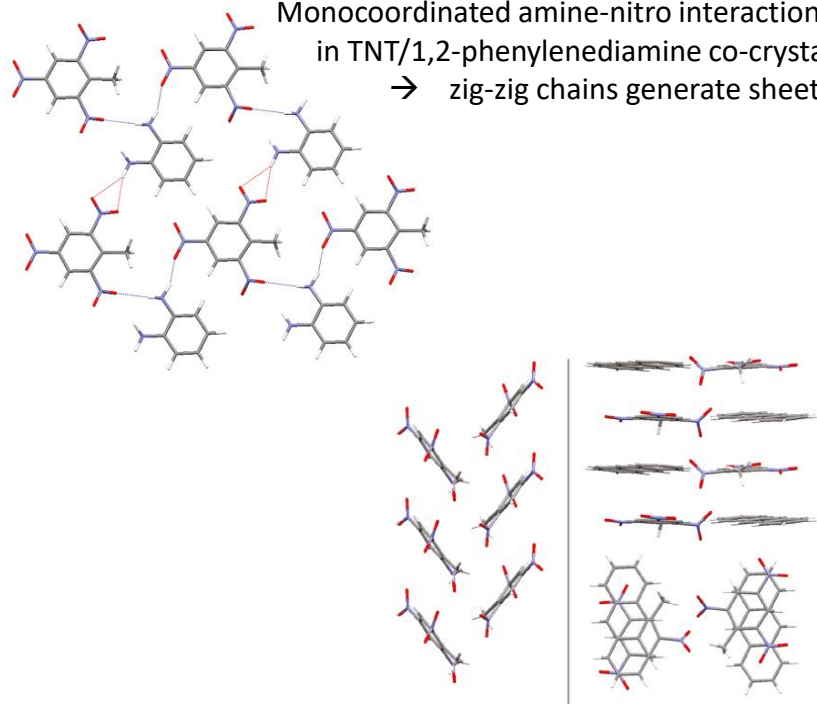
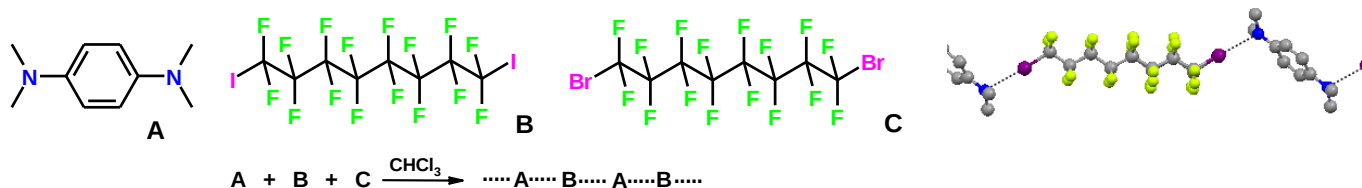
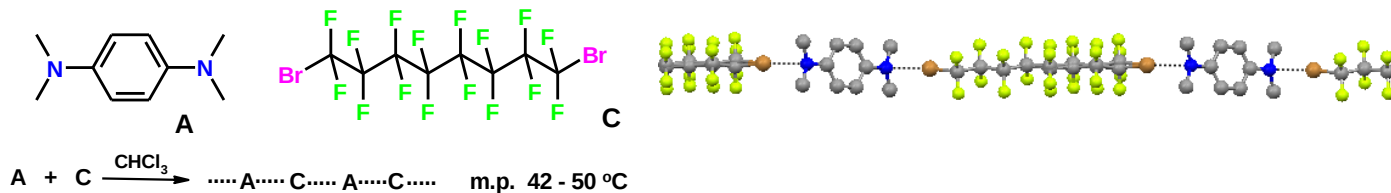
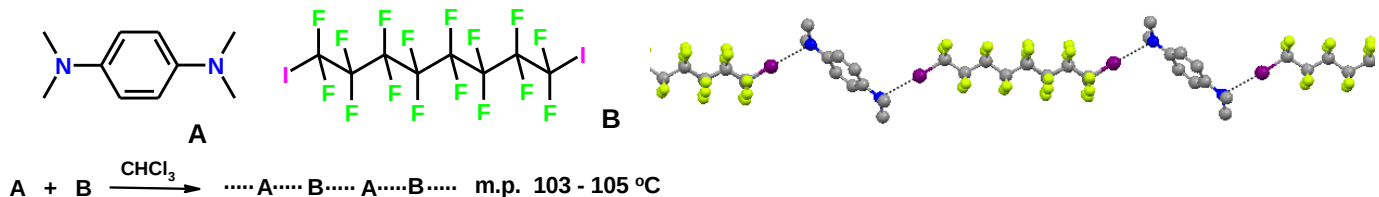


Figure 2. Comparison of the herringbone packing of monoclinic TNT (CSD code ZZZMUC08)³³ (left) and the face-to-face π -stacking of the TNT/anthracene cocrystal (right).

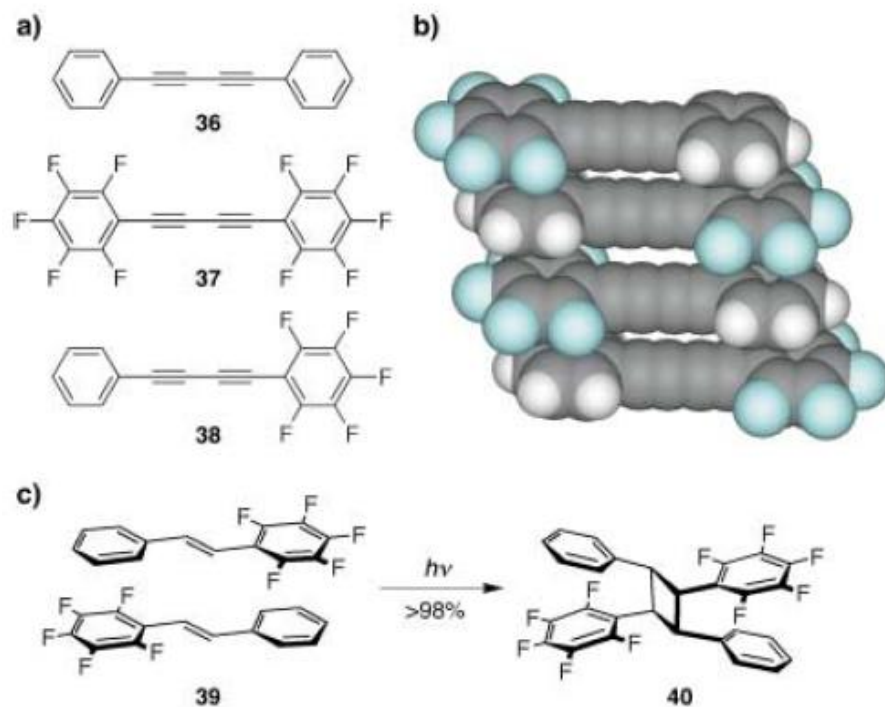
Selectivity in XB donors $\Rightarrow$ competition in interaction strength



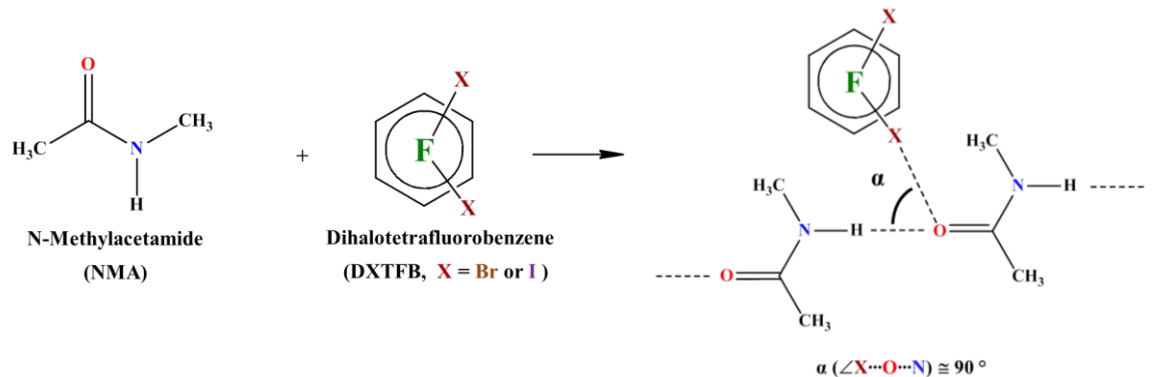
Aryl-perfluoroaryl interactions

- for controlled photoreaction in the solid state

- a) Precursors for cis-polybutadiynes.
b) The X-ray structure of **38** (CSDcode: RIHG0E).
c) The [2+2] photocycloaddition reaction of **39** proceeds with regiochemical and stereochemical control to give **40** crystals.

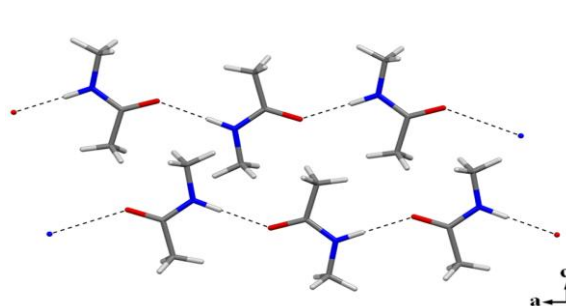


Co-crystals with HB and XB – useful partnership

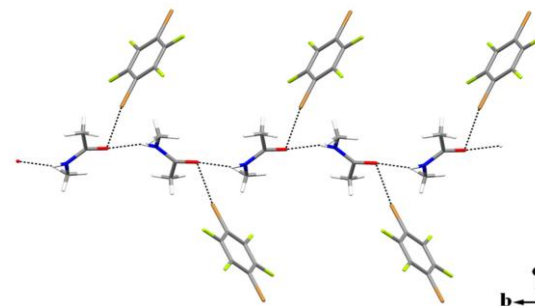


CE of biomimetic systems

- Needed:**
- peptide bond model
 - halogen bond acceptor
 - halogen bond donor



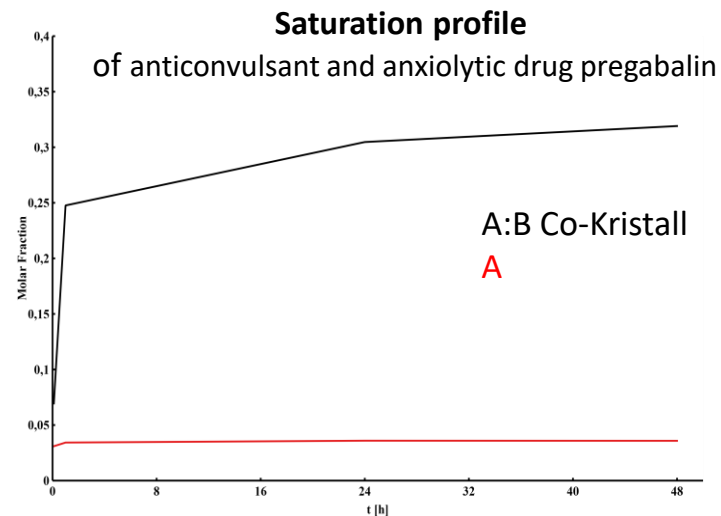
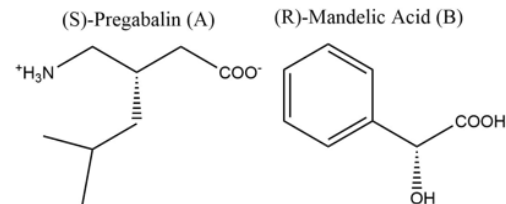
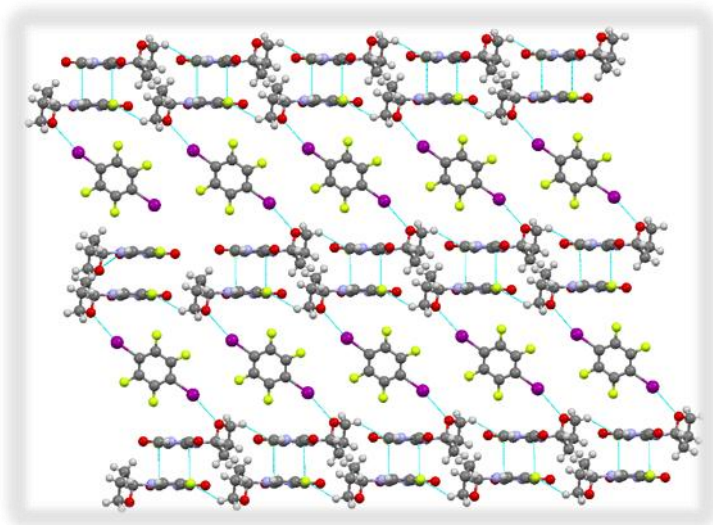
NMA
N-H...O chains



NMA-14DBrTFB
N-H...O chains & Br...O halogen bonds

XB instead of HB in pharmaceutical co-crystals?

Case study of chemotherapist tegafur



Stongly improved solubility in a co-crystal

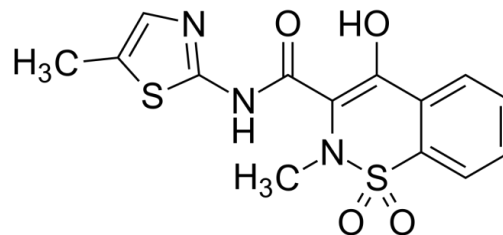
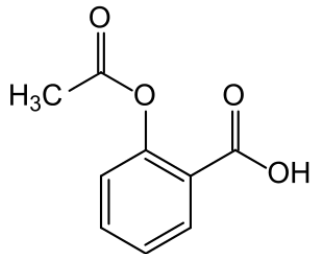
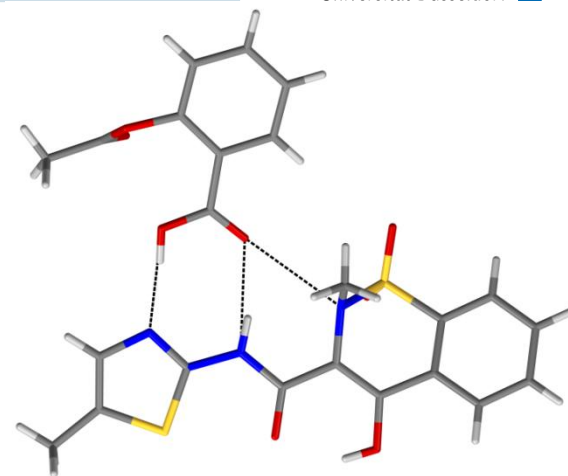
Pharmaceutical co-crystals

aspirin – meloxicam

- analgesic agent and
- non-steroidal antiinflammatory drug

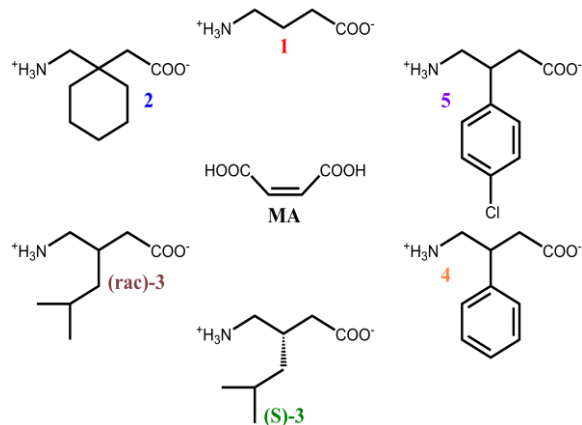
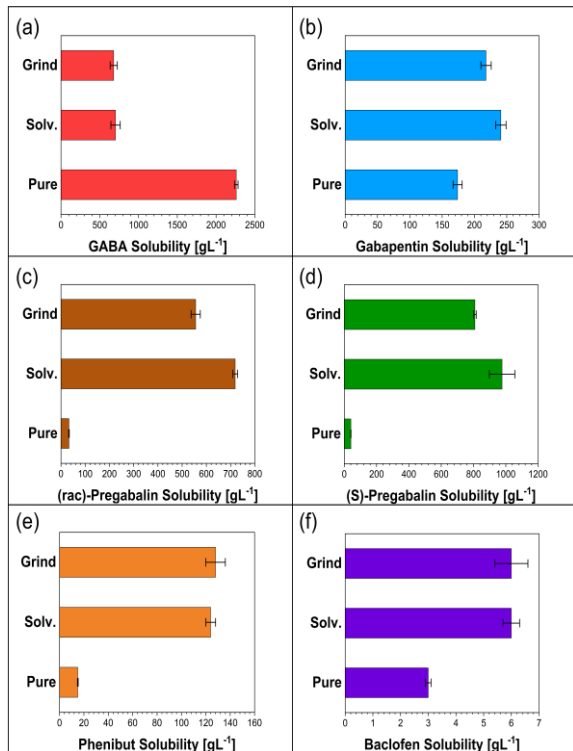
Properties:

- superior kinetic solubility
- improved pharmacokinetic properties
(decreased time for the meloxicam to reach human therapeutic concentration)
- hydrogen bonded dimer

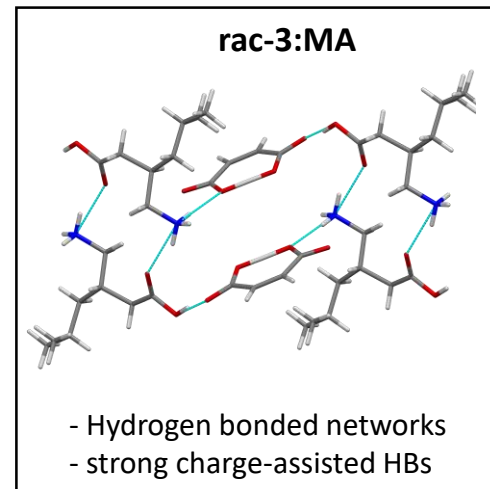


Pharmaceutical co-crystals: systematic analysis

➤ synthetic route and solubility enhancement

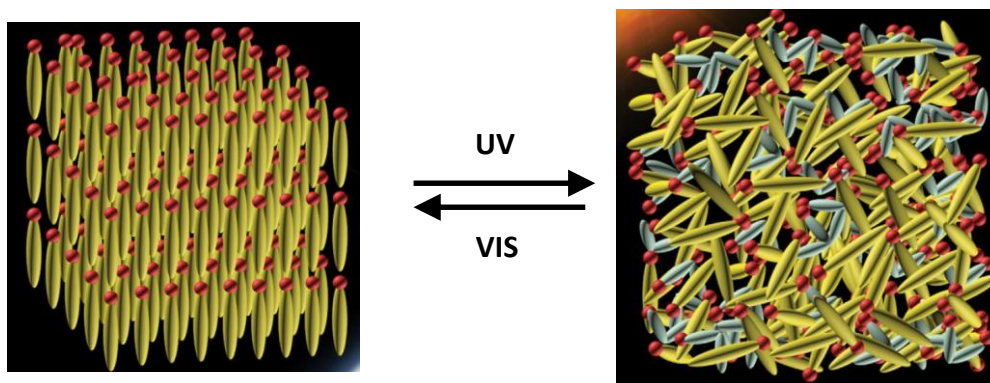
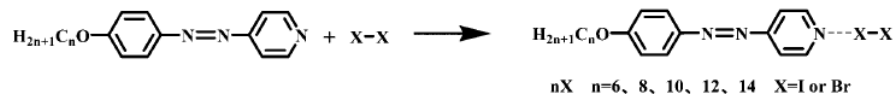


GABA (1), Gabapentin (2), Pregabalin (3), Phenibut (4), Baclofen (5), maleic acid (MA)



- Target APIs: γ -amino butanoic acid derivatives:
- treat epilepsy, neuropathic pain, anxiety disorder ...
- **multicomponent crystals in a *salt-co-crystal continuum***
- **increased solubility up to 23-fold**

Photoresponsive liquid crystals

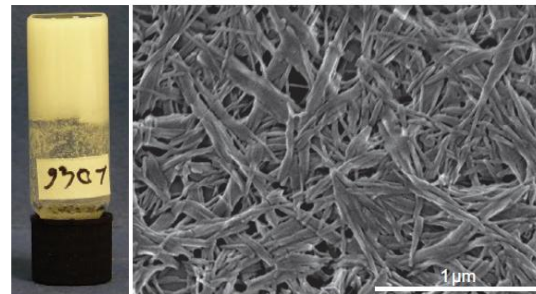
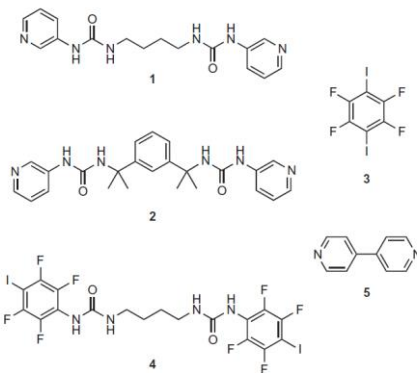
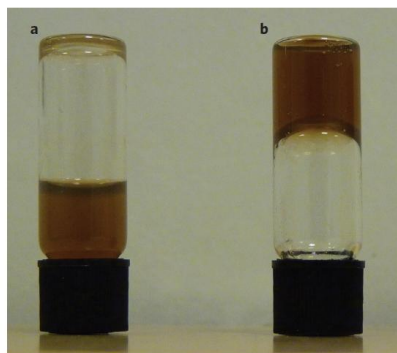


Azopyridines for photocontrolable LC actuator

→ Apps in: flat panel displays, photonics, nanotechnology

Halogen-bonding-triggered supramolecular gel formation

➤ Supramolecular low-molecular-weight gelators:
applications regenerative medicine, drug delivery, controlled crystal growth



Scanning electron micrographs
(SEM) of dried co-gel of **4** with **5**

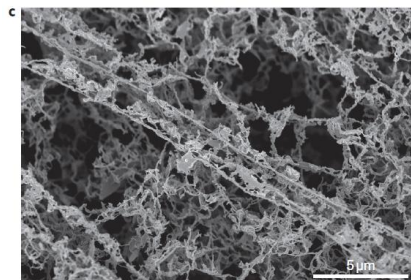
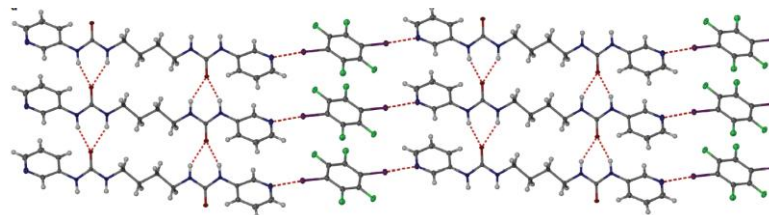
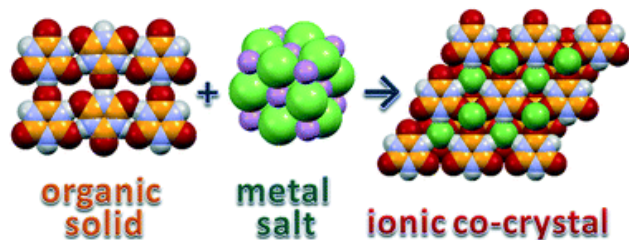


Figure 2 | Chemical formulae of the compounds used to obtain supramolecular LMWGs. The bis(pyridyl urea) gelators **1** and **2** form co-gels in the presence of 1,4-diiodotetrafluorobenzene **3**, and the halogen bond donor bis(urea) derivative **4** forms co-gels when treated with either 4,4'-bipyridine **5**, tetrabutylammonium iodide or pyridyl derivative **1**.

Figure 6 | Co-gel formation between halogen bond acceptor and halogen bond donor bis(urea)s **1** and **4**. **a, b.** Compound **4** in DMSO/water (3:1) does not undergo gelation (**a**), but in the presence of **1** it forms co-gel **1-4** in DMSO/water (3:1) (**b**). **c.** SEM image showing the morphology of the 1-4 co-gel.



ionic co-crystal



Does proton transfer automatically make a salt out of a molecular co-crystal?

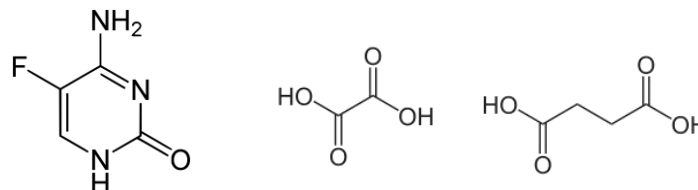
Impact of intermolecular interactions vs. ionicity

- salt-like MCC
- molecular salts
- molecular ionic co-crystals

“salt” vs “co-crystal” formation

ΔpK_a -rule

- rule of thumb - a lot of exceptions!
 - $\Delta pK_a = pK_a(\text{base}) - pK_a(\text{acid})$
 - $\Delta pK_a > 3$ mostly salt
 - $\Delta pK_a < 0$ mostly co-crystal
 - $\Delta pK_a = 0-3$ – salt and co-crystal possible
- ➔ The salt-co-crystal continuum



F-cytosine + oxalic acid ($\Delta pK_s 2$) $\Rightarrow$ salt

F-cytosine + succinic acid ($\Delta pK_s 0.9$) $\Rightarrow$ co-crystal

Potential applications:

- ⇒ optoelectronics & photovoltaics
- ⇒ small ions transporter
- ⇒ functional supramol. liquid crystals
- ⇒ low-weight gels for drug delivery
- ⇒ photoresponsive and luminescent materials
- ⇒ industrial separation
- ⇒ coatings
- ⇒ pharmaceuticals
- ⇒ ...

